

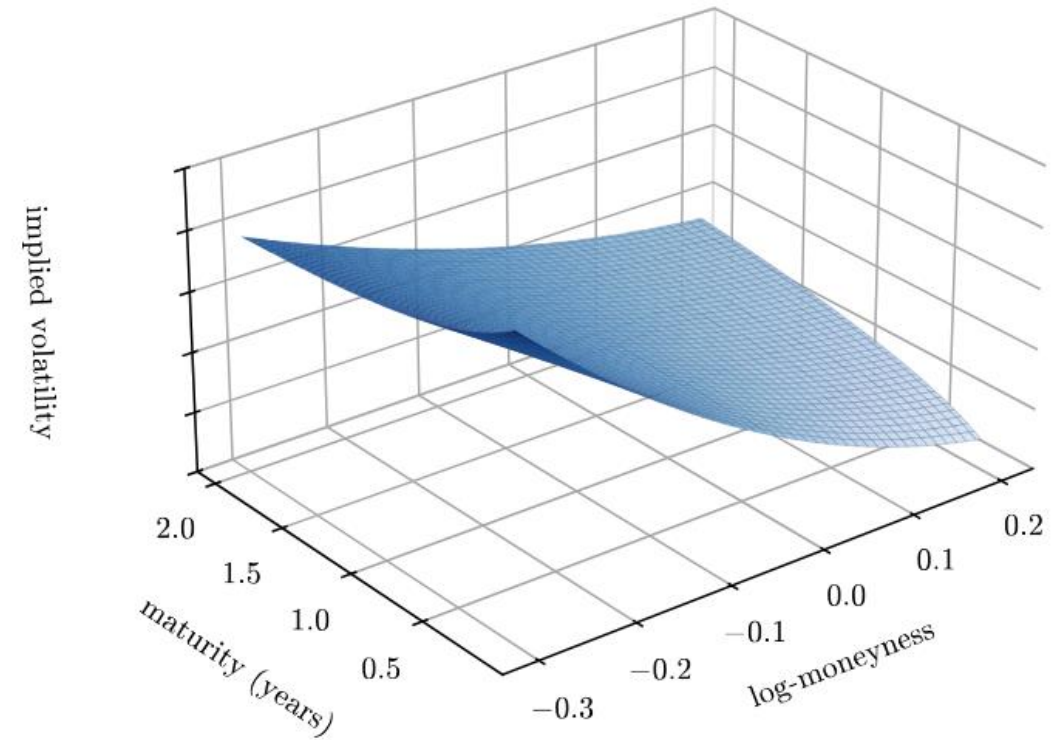
IMPERIAL

High-Dimensional Financial Data Analysis with Tensor Attention-Based Neural Networks

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Forecasting the Implied Volatility Surface

- Options are **insurance against market moves** — their price reveal the market's expected risk.
- Collected across strikes (moneyness levels) and maturities, option quotes form the **implied volatility surface**.
- IV surfaces are primarily used for pricing, hedging and risk management; the latter two depend on how the surface moves, rather than where it sits today.
- Multi-step forecasting of the full IV surface is unaddressed by the literature reviewed.



Research Question:

Is a model that respects the surface's structure better able to forecast the IV surface, multiple steps ahead, from its own history?

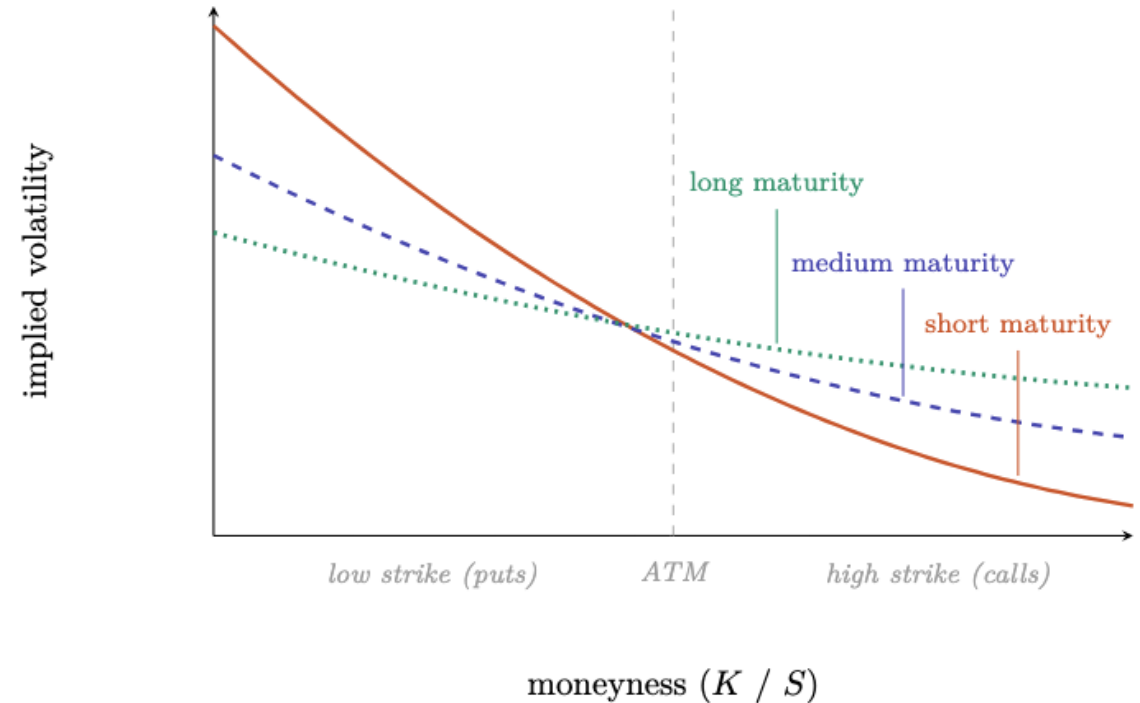
IV Surface Structure

Static structure:

- Equity indices fall faster and harder than they rise → low-strike options (downside protection) are pricier → the **skew**.
- Longer-dated options usually imply higher vol → the **term structure** typically slopes up (and can invert under stress).

Dynamic structure:

- Short-term: **highly persistent** — tomorrow $\approx$ today.
- Most daily evolution is a shared **level shift** (>90% of the variance).
- Longer-term: both level and shape **mean-revert**.



Building the Dataset

The market gives a daily scatter of option quotes — irregular strikes & maturities, changing day to day.

Data	Filtering	Keeping the OTM side
SPX index options (via OptionMetrics), ~ 20 years (2004–2023), ~ 37M quotes	drop unreliable and illiquid quotes (low vega, zero volume)	puts below the forward, calls above — only time value, measured more precisely

Smoothing:

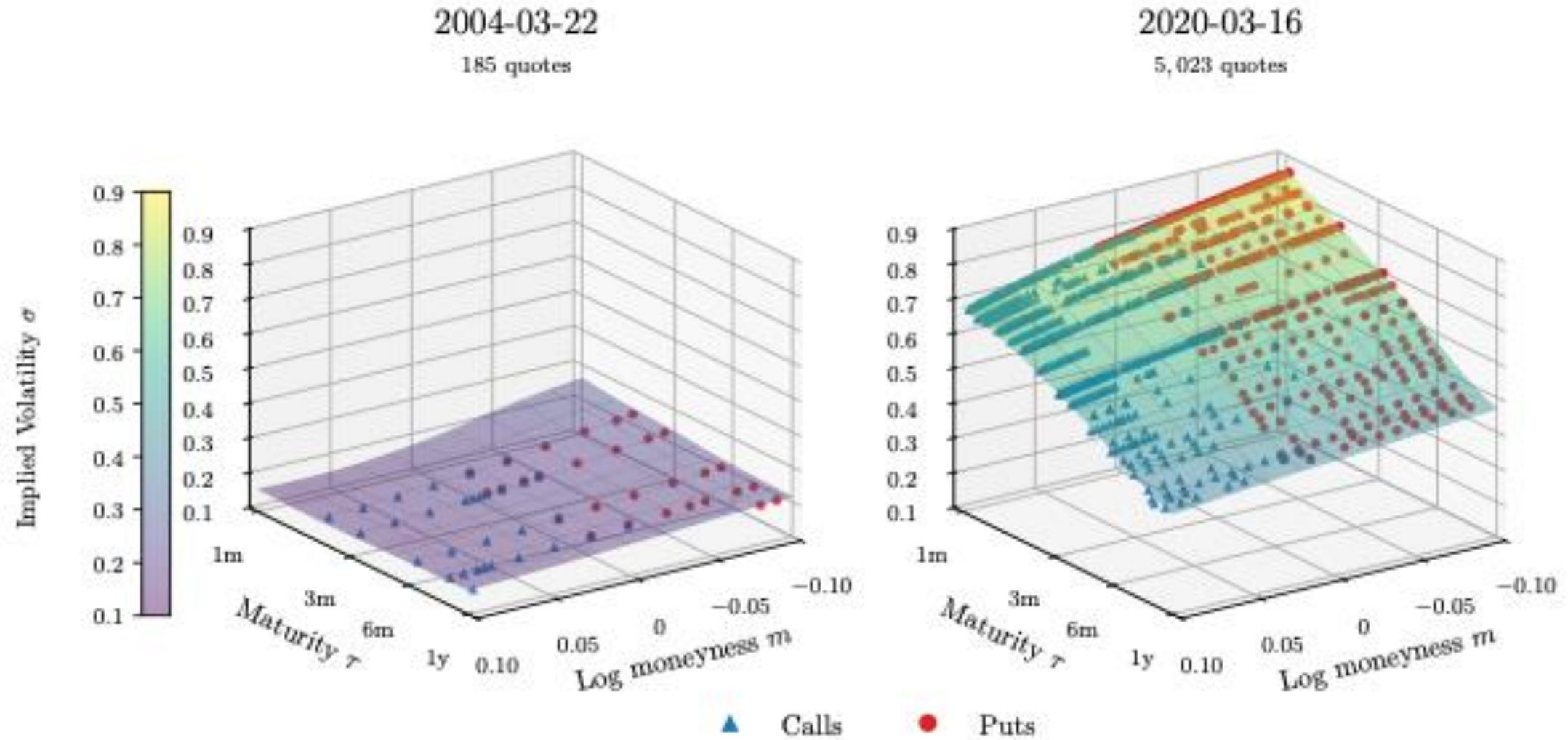
- Vega-weighted Nadaraya-Watson

$$\hat{\sigma}(m, \tau) = \frac{\sum_i \nu_i \sigma_i g(x_i, y_i)}{\sum_i \nu_i g(x_i, y_i)} \quad g(x, y) = \exp\left(-\frac{x^2}{2h_\tau}\right) \exp\left(-\frac{y^2}{2h_m}\right)$$

Resulting surface:

Grid $\rightarrow$ 11 moneyness $(-0.10 \dots + 0.10) \times 10$ maturities $(30 \dots 365 \text{ d}) = \text{one surface/day}$

Representative constructed surfaces



The Forecasting Task

Task:

- Forecasting the full surface (all 110 cells)
- A multi-step path, all steps from base date to horizon.

Data split:

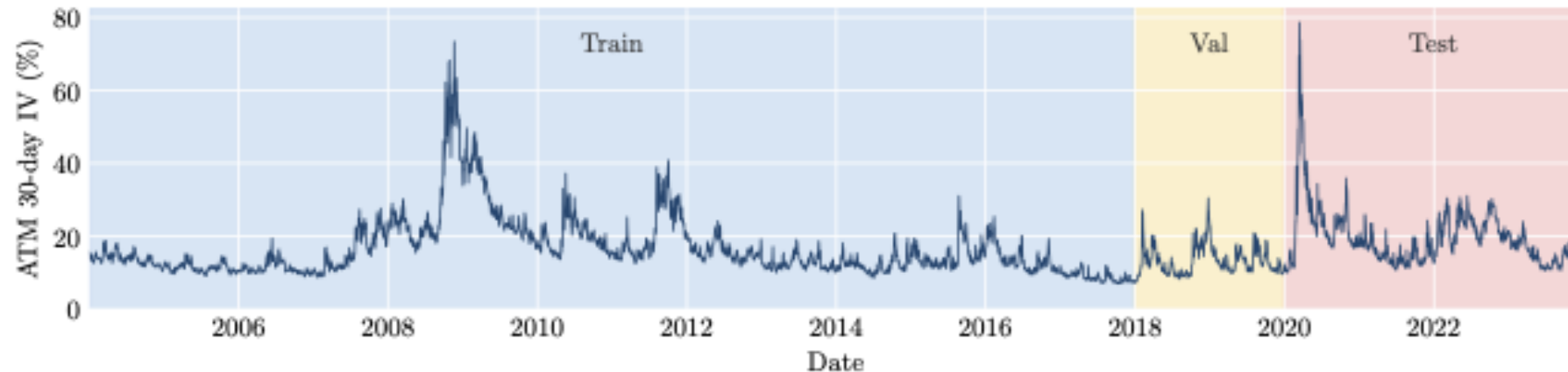
- Chronological 70/10/20 split, with stress episodes present in each set

Horizons:

- $h \in \{10, 21, 42, 63\}$ trading days (approx. 2 weeks, 1 month, 2 months and 1 quarter)

Standardisation:

- Forecasts in standardised log-IV (the literature largely works in log-IV & forecasts are non-negative by default)



Classical Benchmarks

Persistence

- All horizon forecasts are the last seen surface:

$$\hat{z}_{t+i} = z_t \quad \text{for } i = 1, \dots, h, \quad h \in \{10, 21, 42, 63\},$$

- Persistence is the optimal forecast under the assumption that the surface is a martingale.
- IV surfaces are characterised as persistent and near-martingale in short horizons.

Polynomial-Coefficient VAR

- OLS fits the 5-dimensional polynomial on every day in the training set:

$$\log \sigma_t(m, \tau) = \beta_{0,t} + \beta_{1,t}M + \beta_{2,t}M^2 + \beta_{3,t}\tau + \beta_{4,t}(M\tau),$$

where, $M \equiv \frac{m}{\sqrt{\tau}},$

- A vector autoregression is fitted on the training set – lag 2 (BIC).
- In the test set, path forecasts are rolled out step-by-step.

Model Implementation: the Shared Pipeline

Lookback & Centering

- Each model's lookback is 63 days.
- Each model sees deviations to today and predicts deviations from today.
- $\delta = 0$ is persistence; hence the models are 'persistence-anchored'.

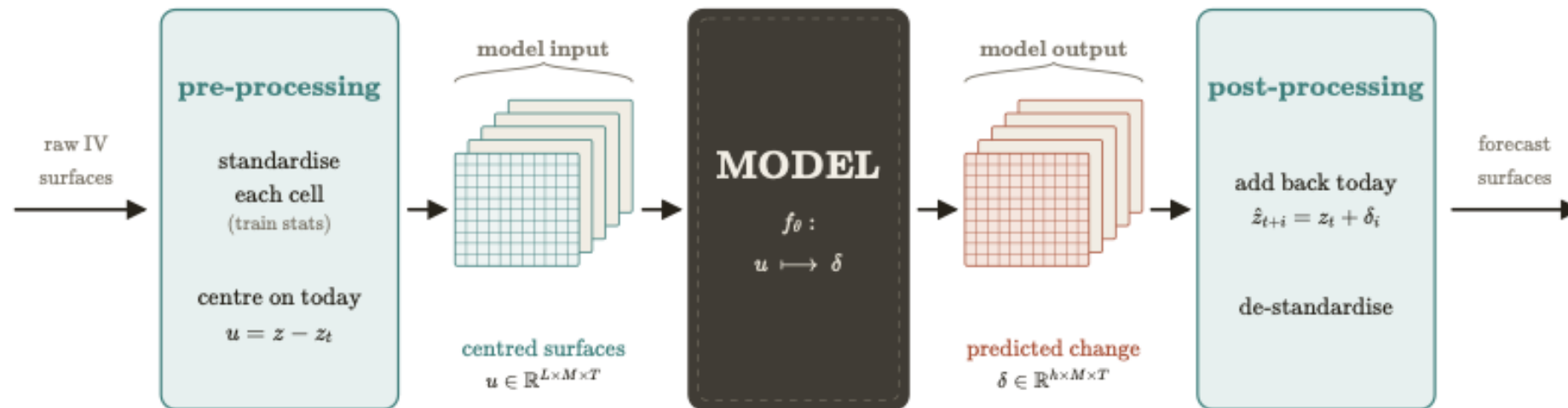
Training

- Each model is trained using the AdamW optimiser via mini-batch SGD.
- The training loss is MSE in standardised log-IV:

$$\mathcal{L}(t, h) = \frac{1}{h \cdot 110} \sum_{i=1}^h \|\hat{z}_{t+i} - z_{t+i}\|_F^2,$$

Parameter Count and Hyperparameters

- All deep models are matched at $\sim 40 - 50k$ parameters.
- Hyperparameters are shared.
- Each model/horizon configuration is trained across 10 random seeds.



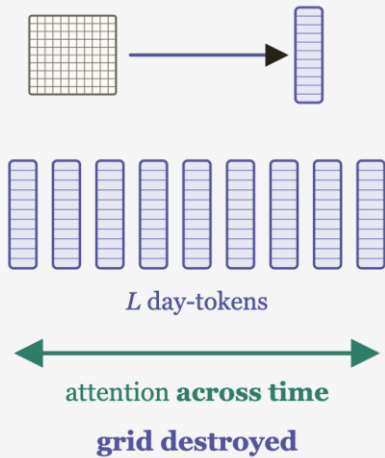
Linear Floor and Plain Transformers

NLinear:

A single linear layer connecting deviations to today and deviations from today. (Identical response in all regimes.)

Day-Token

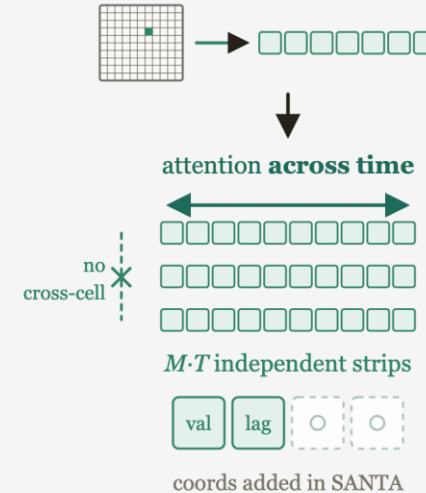
one token = one day



Each day's surface flattened to a single token.

Per-Cell

one token = one cell



Each cell of every day is its own token.

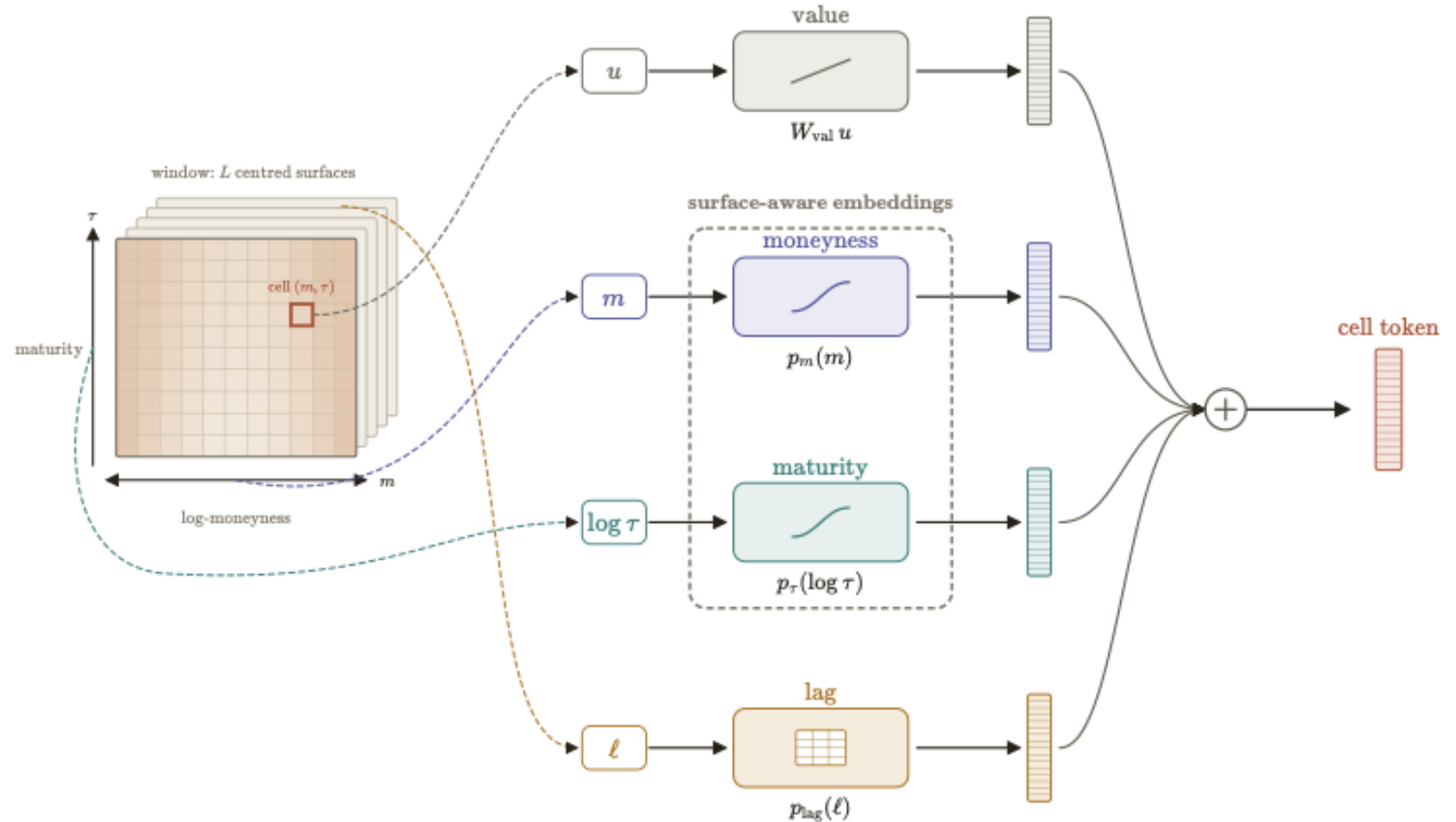
Surface-aware Embeddings

What's added?

- Each cell's moneyness and maturity positions, learned as coordinate embeddings, are summed into its token.

What does this offer?

- The models now know from where on the surface each token comes from, on top of knowing from which lag position.



The surface-aware embeddings remain identical for all SANTA variants.

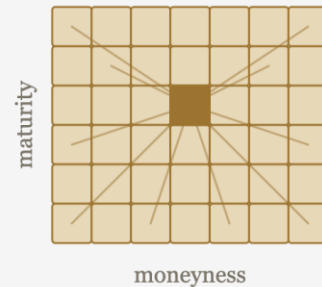
Surface-aware Neural Tensor-attention (SANTA) Variants

SANTA-Temporal:

Identical to the Per-Cell transformer in everything apart from the learned coordinate embeddings.

SANTA-Flat

joint spatial attention



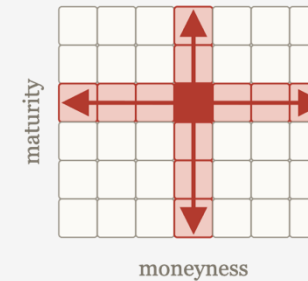
all cells at once

every cell attends to every cell

Unconstrained spatial attention, followed by temporal attention.

SANTA

factored spatial attention



split across axes

moneyness + maturity

Attention across moneyness, then across maturity and finally across time.

Evaluation Metrics

Point-wise Error

MSE:

- The error metric the models were trained to minimise.

$$\text{MSE} = \frac{1}{110 N h} \sum_{n=1}^N \sum_{i=1}^h \sum_{c=1}^{110} (\hat{z}_{n,i,c} - z_{n,i,c})^2$$

MAE:

$$\text{MAE} = \frac{1}{110 N h} \sum_{n=1}^N \sum_{i=1}^h \sum_{c=1}^{110} |\hat{z}_{n,i,c} - z_{n,i,c}|$$

Surface cells are highly correlated — daily changes are mostly a shared level shift.

MSE / MAE mostly score the level: a level miss lands on every cell, so it dominates the summed error.

Cross-sectional Correlation

Cross-sectional Rank IC:

- Forecasted and realised change vs today, per cell:

$$\Delta \hat{z}_c = \hat{z}_c - \text{today}_c, \quad \Delta z_c = z_c - \text{today}_c$$

- Ranking changes of 110 cells, and correlating:

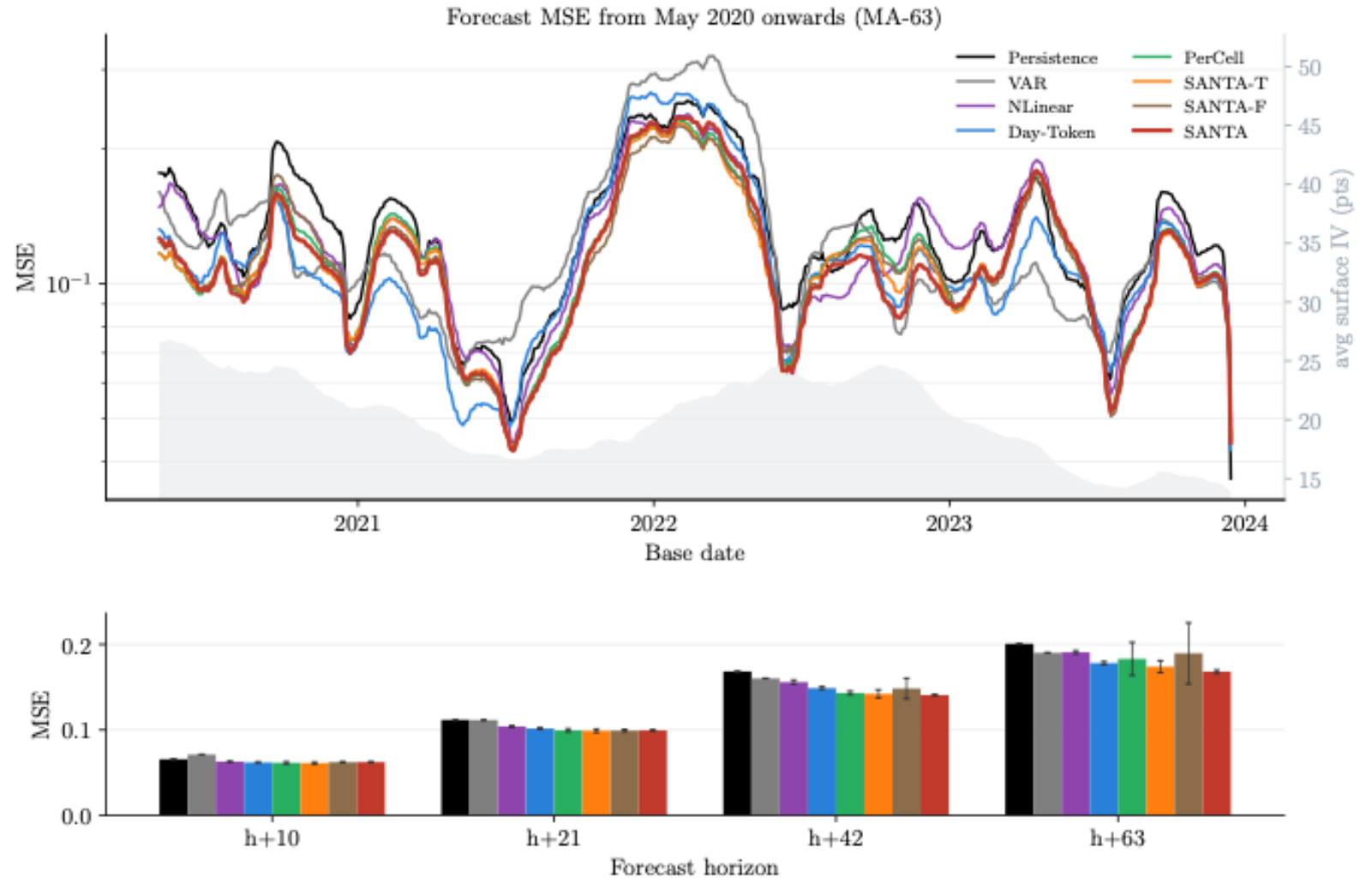
$$\text{IC} = \rho_{\text{Spearman}}(\text{rank}_c \Delta \hat{z}_c, \text{rank}_c \Delta z_c)$$

IC is invariant to a level shift, scoring only the shape: whether a model predicts which cells move most, relative to the rest of the surface.

All reported metrics are calculated in the standardised log-IV space in which the models were trained.

MSE Post-crisis Results

- **Short horizons:** at $h + 10$, persistence is hard to beat.
- **Long horizons:** SANTA records lowest MSE (and MAE) at $h + 42$ and $h + 63$.
- **Stability:** SANTA-Flat's cross-seed std is orders of magnitude larger than SANTA's at $h + 42$ and $h + 63$ – unconstrained spatial attention is less seed-reproducible.
- **Regime pattern:** when the surface level is falling sharply, VAR and Day-Token lead; when it's rising, they trail — they revert confidently toward a fitted long-run level.



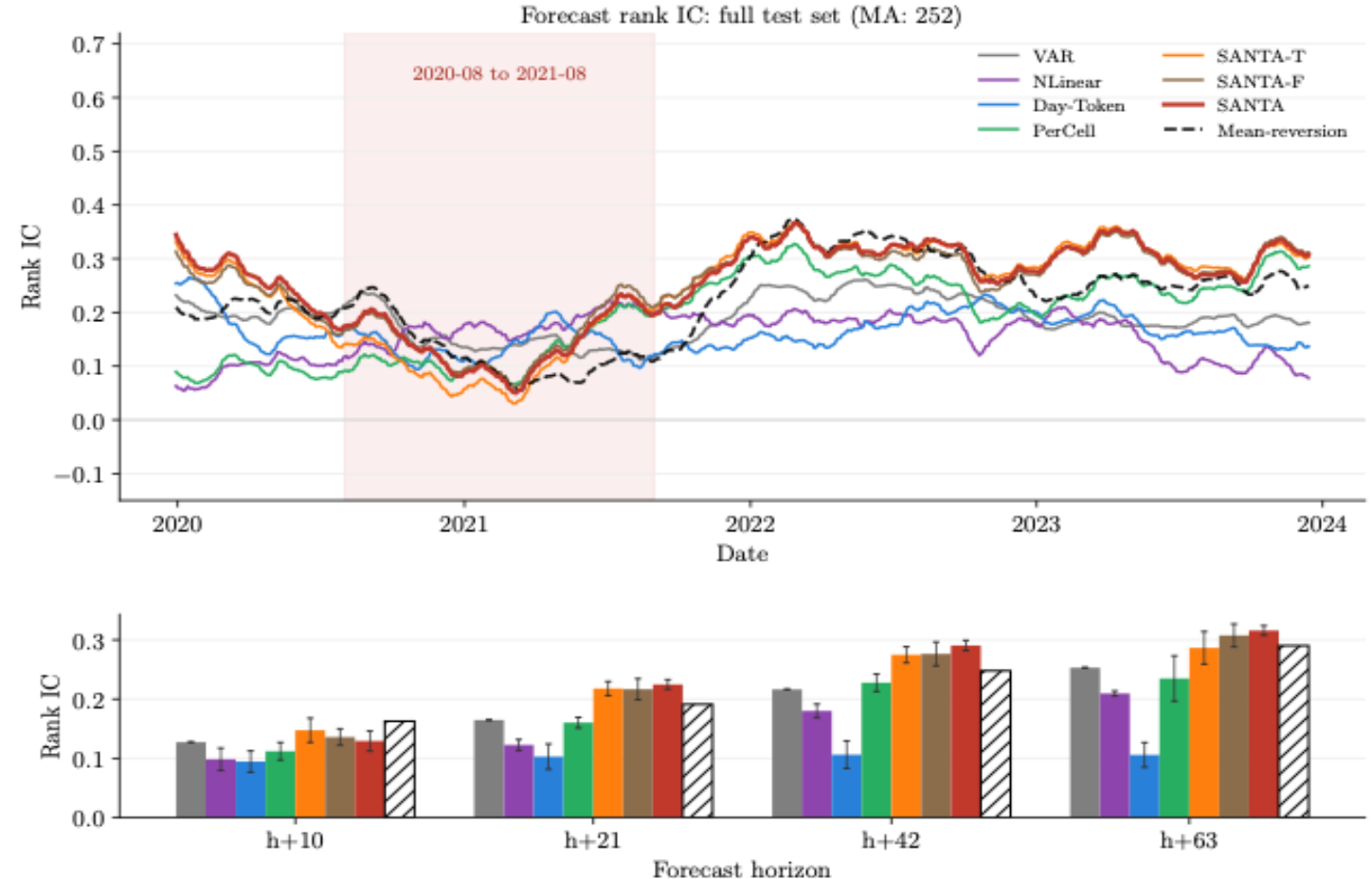
Cross-sectional Rank IC Results

Setup:

- **Reversion rule:** a naïve mean-reversion baseline — each cell predicted to revert toward its long-run average shape.
- **IC rises with horizon:** directional reversion accumulates; random daily shocks partly cancel — so signal outpaces noise further out.

Findings:

- **Short horizons:** at $h + 10$ no model beats the rule.
- **Long horizons:** SANTA records the highest IC from $h + 21$ onwards.
- **Regime caveat:** through 2020–21 the surface's shape stopped reverting to its training-set norm — the rule and the SANTA variants fell together.



Conclusions

Research Question:

Is a model that respects the surface's structure better able to forecast the IV surface, multiple steps ahead, from its own history?

Yes – but selectively.

SANTA is best on every metric at the two longest horizons — uniquely strong on both level and shape.

The ladder isolates which structure earned its place:

- **Anchoring on persistence** — the persistent behaviour is built in; every model beats persistence from the first step.
- **Coordinate embeddings** — carry most of the cross-sectional shape skill, separating the SANTA variants from the field; spatial attention adds little more.
- **Axial attention** — SANTA-Flat's joint attention isn't seed-reproducible; factoring along the two axes restores stability — the better-matched prior

Limitations and Future Work

Limitations

- **One index, one 4-year window** — the findings hold in this period, not a general rule.
- **No hyperparameter tuning** — results are a floor; the ladder's ordering is unproven under a search.
- **No economic evaluation** — the edge is small, and untested against transaction costs.

Future Work

- **Conditioning on external drivers** — forecasts conditioned on exogenous variables (e.g., index returns) could lift performance beyond the history-only setting employed here.
- **Broadening the test** — repeating the study on other underlyings (other equity indices, single-name equities, FX) would test whether the rankings generalise.

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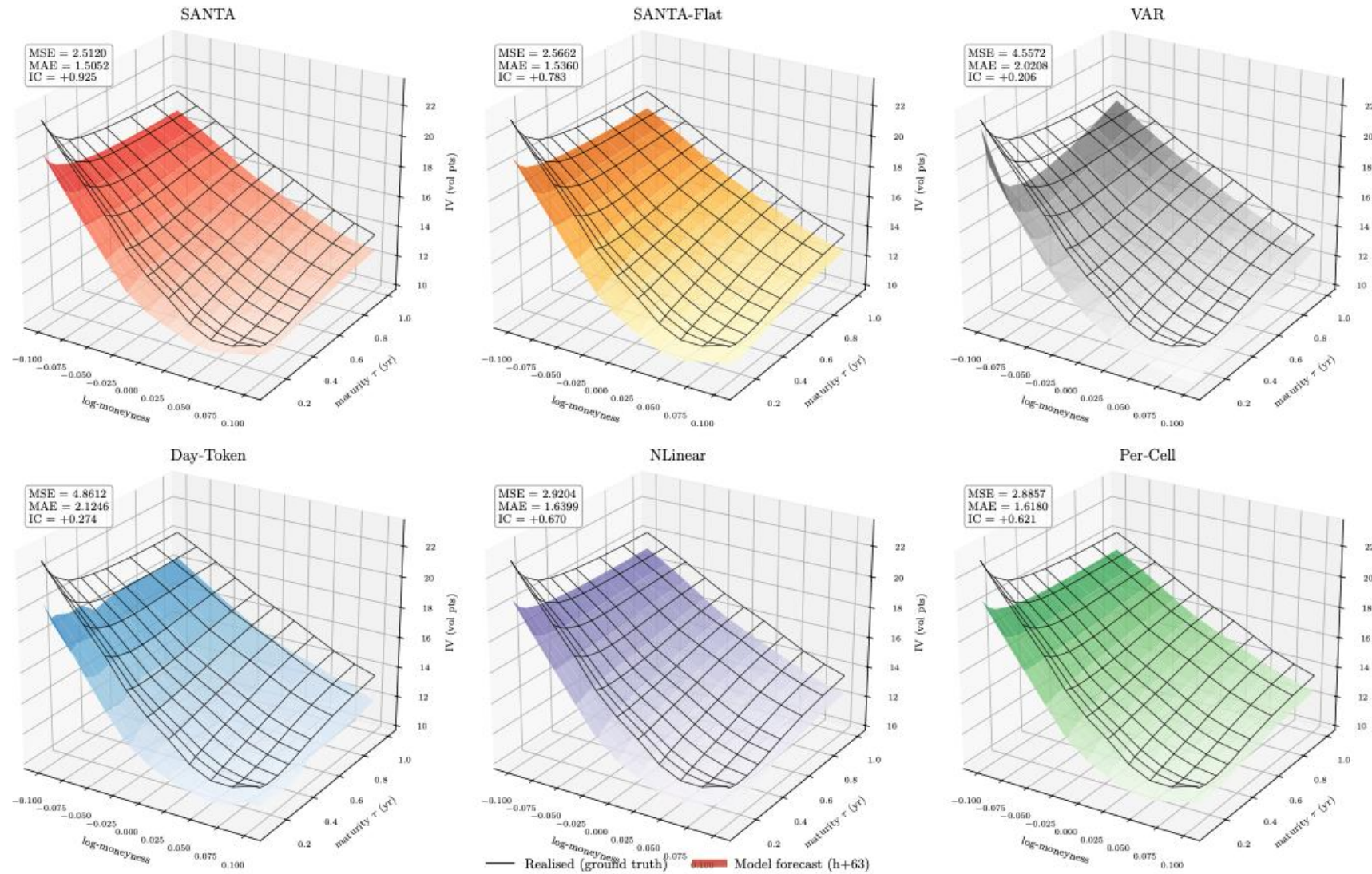
Thank you

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Appendix A: Sample Forecasts

Forecast vs realised IV surface — 2023-10-17 (h+63, seed 0)



Appendix B: In-depth Results Tables

COVID Crisis Period

Horizon	Model	MSE	MAE	IC
$h+10$	Persistence	0.57205	0.49506	—
	VAR	0.63519	0.49722	0.2587
	NLinear	0.63289 ± 0.0216	0.49596 ± 0.0149	0.1054 ± 0.042
	Day-Token	0.61487 ± 0.0055	0.49613 ± 0.0041	0.1175 ± 0.027
	Per-Cell	0.60346 ± 0.0081	0.50022 ± 0.0050	-0.0091 ± 0.035
	SANTA-T	0.59774 ± 0.0154	0.49873 ± 0.0103	0.1537 ± 0.103
	SANTA-F	0.58432 ± 0.0070	0.48927 ± 0.0035	0.0720 ± 0.133
	SANTA	0.57953 ± 0.0070	0.48816 ± 0.0038	0.0815 ± 0.052
$h+21$	Persistence	1.41394	0.77147	—
	VAR	1.52364	0.74471	0.2790
	NLinear	1.45395 ± 0.0202	0.74399 ± 0.0121	-0.0022 ± 0.044
	Day-Token	1.44932 ± 0.0201	0.74987 ± 0.0069	0.1452 ± 0.018
	Per-Cell	1.40200 ± 0.0119	0.74327 ± 0.0061	-0.0587 ± 0.048
	SANTA-T	1.39602 ± 0.0371	0.74102 ± 0.0222	0.3878 ± 0.102
	SANTA-F	1.39093 ± 0.0184	0.73747 ± 0.0084	0.3340 ± 0.120
	SANTA	1.38022 ± 0.0126	0.73365 ± 0.0057	0.3807 ± 0.021
$h+42$	Persistence	2.92332	1.23002	—
	VAR	2.99228	1.14830	0.2987
	NLinear	2.79960 ± 0.0378	1.13972 ± 0.0226	0.0433 ± 0.056
	Day-Token	2.86993 ± 0.0231	1.16687 ± 0.0100	0.1628 ± 0.033
	Per-Cell	2.73979 ± 0.0297	1.14025 ± 0.0074	0.0575 ± 0.080
	SANTA-T	2.72616 ± 0.0596	1.13467 ± 0.0163	0.4839 ± 0.038
	SANTA-F	2.83230 ± 0.1416	1.17895 ± 0.0523	0.4300 ± 0.053
	SANTA	2.68792 ± 0.0297	1.12831 ± 0.0099	0.4821 ± 0.054
$h+63$	Persistence	3.59735	1.45290	—
	VAR	3.54198	1.31979	0.3578
	NLinear	3.30643 ± 0.0355	1.29659 ± 0.0240	0.0994 ± 0.059
	Day-Token	3.45039 ± 0.0409	1.35475 ± 0.0259	0.1916 ± 0.031
	Per-Cell	3.22360 ± 0.0453	1.30209 ± 0.0114	0.1868 ± 0.079
	SANTA-T	3.20419 ± 0.0461	1.28955 ± 0.0168	0.5177 ± 0.036
	SANTA-F	3.23018 ± 0.1004	1.30458 ± 0.0350	0.5039 ± 0.067
	SANTA	3.17829 ± 0.0736	1.29655 ± 0.0156	0.5457 ± 0.065

Post-crisis Period

Horizon	Model	MSE	MAE	IC
$h+10$	Persistence	0.06566	0.18521	—
	VAR	0.07116	0.19731	0.1151
	NLinear	0.06284 ± 0.0007	0.18573 ± 0.0011	0.0977 ± 0.018
	Day-Token	0.06188 ± 0.0007	0.17995 ± 0.0008	0.0924 ± 0.019
	Per-Cell	0.06150 ± 0.0013	0.18088 ± 0.0022	0.1232 ± 0.015
	SANTA-T	0.06117 ± 0.0012	0.18088 ± 0.0013	0.1468 ± 0.016
	SANTA-F	0.06226 ± 0.0007	0.18204 ± 0.0011	0.1419 ± 0.010
	SANTA	0.06237 ± 0.0006	0.18110 ± 0.0007	0.1337 ± 0.019
$h+21$	Persistence	0.11166	0.24921	—
	VAR	0.11146	0.25183	0.1539
	NLinear	0.10395 ± 0.0010	0.24502 ± 0.0010	0.1344 ± 0.010
	Day-Token	0.10162 ± 0.0009	0.23845 ± 0.0014	0.0989 ± 0.023
	Per-Cell	0.09942 ± 0.0019	0.23801 ± 0.0025	0.1809 ± 0.010
	SANTA-T	0.09896 ± 0.0020	0.23872 ± 0.0025	0.2018 ± 0.008
	SANTA-F	0.09916 ± 0.0015	0.23831 ± 0.0018	0.2057 ± 0.009
	SANTA	0.09942 ± 0.0008	0.23835 ± 0.0011	0.2100 ± 0.008
$h+42$	Persistence	0.16828	0.31537	—
	VAR	0.16021	0.30855	0.2085
	NLinear	0.15584 ± 0.0024	0.30676 ± 0.0016	0.1932 ± 0.014
	Day-Token	0.14889 ± 0.0017	0.29566 ± 0.0016	0.1007 ± 0.023
	Per-Cell	0.14320 ± 0.0021	0.29276 ± 0.0019	0.2441 ± 0.013
	SANTA-T	0.14218 ± 0.0047	0.29282 ± 0.0049	0.2546 ± 0.014
	SANTA-F	0.14842 ± 0.0119	0.29634 ± 0.0093	0.2615 ± 0.021
	SANTA	0.14074 ± 0.0008	0.29063 ± 0.0009	0.2720 ± 0.010
$h+63$	Persistence	0.20062	0.34791	—
	VAR	0.19014	0.33704	0.2427
	NLinear	0.19078 ± 0.0020	0.34335 ± 0.0017	0.2204 ± 0.007
	Day-Token	0.17810 ± 0.0018	0.32548 ± 0.0021	0.0973 ± 0.021
	Per-Cell	0.18303 ± 0.0193	0.33139 ± 0.0147	0.2395 ± 0.039
	SANTA-T	0.17392 ± 0.0070	0.32527 ± 0.0060	0.2637 ± 0.029
	SANTA-F	0.18954 ± 0.0356	0.33233 ± 0.0223	0.2880 ± 0.015
	SANTA	0.16806 ± 0.0020	0.32008 ± 0.0024	0.2930 ± 0.013